

Course Information Sheet

Second-year Mathematics at UCT in 2023

You should read this document if you are registered for any of the following courses:

- MAM2000W
- MAM2002S
- MAM2003Z
- MAM2004H

1 Course structure

The second-year mathematics courses at UCT are modularised. Each module consists, nominally, of 30 lectures, and depending on which course you have registered for you will attend lectures and tutorial, and write class tests and exams for either four (MAM2000W), two (MAM2002S, MAM2004H), or one (MAM2003Z) module(s).

Each module has its own module leader. Module leaders will provide module-specific details about teaching and learning activities, including assessments. If you have issues or queries specific to a module you should contact the module leader.

If you have an issue or query about the course, you should contact the course convener, Mr Thomas van Heerden (thomas.vanheerden@uct.ac.za), but please read this document and the announcements on Vula carefully before you do. I've tried to make all the information you need available to you. If you still do need to email me put the course you are registered for in the subject line e.g. "MAM2004H Question about final mark".

1.1 Advanced Calculus (2AC)

- **Module Leader:** Dr Francois Ebobisse-Bille (francois.ebobissebille@uct.ac.za)
- **Additional Lecturer:** Ms Sané Erasmus (sane.erasmus@uct.ac.za)
- **Syllabus:** Multivariable calculus. Curves and surfaces in three dimensions, change of coordinates. Line integrals, surface integrals. Stokes', Green's and divergence theorems.
- **First Semester Module**

1.2 Linear Algebra (2LA)

- **Module Leader:** Dr Holger Spakowski (holger.spakowski@uct.ac.za)
- **Additional Lecturer:** Mr Simon Chili (simon.chili@uct.ac.za)
- **Syllabus:** Vector spaces, linear independence, spans, bases, row space, column space, null space. Linear maps. Eigenvectors and eigenvalues. Inner product spaces, orthogonality.
- **First Semester Module**

1.3 Differential Equations (2DE)

- **Module Leader:** Mr Thomas van Heerden (thomas.vanheerden@uct.ac.za)
- **Syllabus:** Linear ordinary and partial differential equations and systems of differential equations.
- **Second Semester Module**
- **Prerequisites:** 2AC and 2LA.
- *Students who have a credit for an Applied Mathematics course containing the modules 2OD and/or 2BP will not be credited for a course containing the module 2DE.*

1.4 Introductory Algebra (2IA)

- **Module Leader:** Dr Juana Sanchez-Ortega (juana.sanchez-ortega@uct.ac.za)
- **Syllabus:** Induction, strong induction and Well-Ordering axiom. Divisibility and prime factorization. Modular Arithmetic. Permutations. Groups. Subgroups. Cyclic groups. Isomorphisms. Simple groups. Factor groups. Lagrange's Theorem. The First Isomorphism Theorem.
- **Second Semester Module**

1.5 Real Analysis (2RA)

- **Module Leader:** Dr Elena Berdysheva (elena.berdysheva@uct.ac.za)
- **Additional Lecturer:** Dr Francesco Russo (francesco.russo@uct.ac.za)
- **Syllabus:** Axioms of the real numbers, supremum and infimum. Countable sets. Sequences and series. Open and closed sets, compactness. Limits, continuity, differentiability. Sequences and series of functions, uniform convergence, power series.
- **Second Semester Module**

2 Module Selection and Progression

All students should complete the online module selection form by 21 February. You can access the form at <https://forms.gle/jevx7sRk3kr6ZAwe6>. This form is not binding. The real determinant of which modules you are registered for is which sets of tests and exams you complete. This form is to give us more information in advance of those tests and exams so that we can make suitable administrative arrangements.

2.1 Which modules should you choose?

- If you are taking MAM2000W you must do two first-semester modules and two second-semester modules. Since there are only two first-semester modules you must take 2AC and 2LA. In the second semester you can choose two of the three that are offered.
 - If you want to take MAM3000W next year, take 2RA and 2IA since those are prerequisites for the third-year modules.
 - If you are majoring in Applied Mathematics you cannot take 2DE (it overlaps too much with material you've already seen), so take 2RA and 2IA.
 - If you are an Actuarial Science student take 2RA and 2DE.

- If you have previously passed any of MAM2002S, MAM2003Z or MAM2004H you cannot repeat the same modules for which you've already claimed credit.

For example: last year Andrew passed MAM2004H taking the modules 2AC and 2RA. They also failed 2LA in the first semester last year. This year they register for MAM2002S, and since they want to do MAM3000W next year, they take the modules 2LA (repeating the failed module) and 2IA. If they were an actuarial science student they would instead take 2LA and 2DE.

If you sign up for modules that you've already been awarded credit for, you will be deregistered from the course. Please don't do this since it creates an enormous amount of work for us, and will delay your degree by at least a year.

3 Assessment, Marks, DP and Progression

Each module that you take is assessed separately, so if you are registered for MAM2000W you will have four class record marks and four exam marks, one for each module.

The composition of the class record (CR) for each module will be determined by the module leaders. Please speak to your module leader for more information. In addition, each module will have a final exam. Your final mark for each module is calculated from your CR and final exam mark (EX) using

$$\text{Final Module Mark} = \max \{40\%CR + 60\%EX, 20\%CR + 80\%EX\}.$$

Your final mark for the course is the average of the Final Module Marks you scored for each module in your course. Please note that there is a difference between taking MAM2000W and MAM2002S & MAM2004H. In the first case four of your Final Module Marks are averaged, and in the second there are two sets of two averaged Final Module Marks. It is not possible to combine the marks from different courses.

If you miss a test for good reason, please email your module leader with an explanation (illness, death in the family, mugging, etc.) along with some evidence (medical certificate, death certificate, case number, etc.) as soon as possible. That email should be copied to me (Thomas) as well. We appreciate that this may not be the day after the test if you are seriously ill, and that's okay.

Please note that departmental policy is to accept medical certificates that state "According to my knowledge" and not to accept certificates that state "As I was informed". Please ask your practitioner to be clear on the certificate.

If you miss a test the module leader may require you to be reassessed. The form of that reassessment will be determined by the module leader. Alternatively the remaining assessments may be reweighted to produce your Final Module Mark. If you miss all the assessments for a module you will have to sit an alternative assessment before the exam. This is normally an oral examination and your module leader will liaise with you to arrange a mutually suitable time.

If you write a test the mark that you achieve is final. We cannot retroactively excuse you from the test. If there is a good reason why the test isn't going to be an accurate reflection of your ability (e.g. illness, death in the family etc.) then don't write it and send us the email with an explanation of your situation. If you do write the test we will treat the mark as a fair indicator of your progress.

The DP requirement for the course is a class record of 30%, and this is checked independently for each module, i.e. you need a class record of 30% for 2AC to be allowed to write the 2AC exam, a class record of 30% for 2LA to be allowed to write the 2LA exam and so on. Note that even if the average of all your class records is above 30% you won't be allowed to write the exams for the modules where your class record is below 30%.

Some of the second-semester modules have prerequisites. In all cases you are required to score at least 45% for the prerequisite module to continue with the appropriate second-semester module. No concessions for prerequisites will be granted.

- 2RA has no prerequisites.
- 2IA has no prerequisites.
- 2DE requires 2AC and 2LA.

In addition, you must score at least 45% for both first-semester modules if you want to continue with MAM2000W in the second semester. If you fail to meet these requirements your registration will be changed and you may still be able to claim credit for MAM2002S. Some examples:

- Bongi registers for MAM2000W, taking 2AC, 2LA, 2RA and 2IA. They pass 2AC and 2LA in the first semester and continues with 2RA and 2IA in the second semester.
- Charles registers for MAM2000W, taking 2AC, 2LA, 2RA and 2IA. They pass 2AC, but only score 40% for 2LA. They are deregistered for MAM2000W, and registered for MAM2002S. MAM2002S requires two modules, so they take 2RA (the only second semester module he qualifies for).

Charles passes 2RA, and the average of 2AC and 2RA is the final grade for MAM2002S. The following year they register for MAM2004H, this time taking 2LA and 2IA.

- Dawn registers for MAM2000W, taking 2AC, 2LA, 2RA and 2IA. They score 40% for 2AC and 35% for 2LA. They are deregistered from MAM2000W, and registered for MAM2004H which appears as a fail on her transcript.

The following year they can register for MAM2000W again.

These changes in your registration, along with technical fee waivers (if you are changed from a W course to an H or S+Z course), will be processed automatically by the Faculty Office. You don't need to fill in any forms.

For MAM2000W students: At the end of the year if the average of all four of your modules is less than 50% your mark is recorded against MAM2000W as a fail. However, if there is a combination of two modules that averages to 50% or better then you will have MAM2002S added to your transcript with the CX (Credit Exemption) symbol. This is equivalent to a pass in MAM2002S so the following year you will register for MAM2004H and only have to repeat the outstanding modules. For example

- Ezra registers for MAM2000W and scores 45% for 2AC and 50% for 2LA in the first semester. In the second semester they score 35% for 2RA and 50% for 2IA. The average of these four modules is 45%, so they fail MAM2000W.

However, the average of 2LA and 2IA is 50%, so they receive an exemption for MAM2002S and next year only have to repeat 2AC and 2RA.

4 Lectures and Tutorials

First semester lectures take place in M304 and M320 every day during fifth period (12:00 - 13:00). The lectures alternate between 2AC and 2LA according to the following schedule.

- **2LA:** Mondays, Thursdays, some Wednesdays (Feb 15; Mar 1, 15; Apr 12, 26; May 3, 10)
- **2AC:** Tuesdays, Fridays, some Wednesdays (Feb 22; Mar 8, 22; Apr 5, 19; May 17)

We will post second semester lecture information to Vula later in the year.